

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Microsoft TRP3 (SAT46), TX -- station KSSF, America/Chicago
Committed pair OSM 1010260677 -> 1533350872
 Microsoft / Microsoft
Facade gap 136.3 m between the two halls
Weather record 43,628 real hourly records from KSSF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.7003 C at
 200 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets
and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs.
Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was
generated at build time for the ONE configuration named below, chosen because it is the configuration
in which this site's agent does the most while still changing mode at least once. If the numbers on
screen differ from the numbers here, the configuration differs -- compare the two lists before
concluding anything else.

Day 2021-01-05 (safe_sector)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 14 of 24 hours with 2 mode changes. That is 4 more than the reactive
incumbent, at 0 unsafe hours against its 0. 6 hours were safe but still ran chillers, because the
schedule could not afford them.

Agent 14 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
 6 hour(s) passed every gate and still ran chillers, because the schedule could not afford
 them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	12.780	18.0	12.780	1.130
01	FREE	yes	-	11.945	18.0	11.110	0.934
02	FREE	yes	-	11.945	18.0	11.110	0.760
03	FREE	yes	-	11.670	18.0	10.560	0.598
04	FREE	yes	-	11.110	18.0	10.560	0.778
05	FREE	yes	-	10.826	18.0	9.711	0.683
06	FREE	yes	-	10.005	18.0	8.890	0.643
07	FREE	yes	-	10.875	18.0	10.000	0.563
08	FREE	yes	-	12.225	18.0	10.560	1.267
09	FREE	yes	-	14.445	18.0	11.670	2.251
10	FREE	yes	-	16.390	18.0	12.780	2.474
11	FREE	yes	-	16.390	18.0	14.440	1.473
12	FREE	yes	-	16.671	18.0	15.000	1.137
13	FREE	yes	-	17.780	18.0	16.670	1.173
14	mechanical	no	dry-bulb	18.330	18.0	17.220	1.004
15	mechanical	no	dry-bulb	18.610	18.0	18.330	0.914
16	mechanical	no	dry-bulb	18.890	18.0	18.330	0.931
17	mechanical	no	dry-bulb	18.050	18.0	17.220	1.134
18	mechanical	yes	switch budget	17.501	18.0	16.670	1.112

19	mechanical	yes	switch budget	16.945	18.0	16.670	1.277
20	mechanical	yes	switch budget	16.562	18.0	16.670	1.546
21	mechanical	yes	switch budget	16.116	18.0	16.670	1.405
22	mechanical	yes	switch budget	16.391	18.0	16.670	1.329
23	mechanical	yes	switch budget	16.395	18.0	16.670	1.037

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.780 C, which is 5.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.130 C, and the margin is measured, not chosen: 1.130 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.945 C, which is 6.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.934 C, and the margin is measured, not chosen: 0.933 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.945 C, which is 6.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.761 C, and the margin is measured, not chosen: 0.760 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.598 C, and the margin is measured, not chosen: 0.598 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.110 C, which is 6.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.778 C, and the margin is measured, not chosen: 0.777 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.826 C, which is 7.174 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.683 C, and the margin is measured, not chosen: 0.682 C of group-conditional forecast error for this hour of day, plus 0.0006 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.005 C, which is 7.995 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.643 C, and the margin is measured, not chosen: 0.642 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.875 C, which is 7.125 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.563 C, and the margin is measured, not chosen: 0.562 C of group-conditional forecast error for this hour of day, plus 0.0006 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.225 C, which is 5.775 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.267 C, and the margin is measured, not chosen: 1.267 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how

much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.445 C, which is 3.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.251 C, and the margin is measured, not chosen: 2.251 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.390 C, which is 1.610 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.474 C, and the margin is measured, not chosen: 2.474 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.390 C, which is 1.610 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.473 C, and the margin is measured, not chosen: 1.473 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.670 C, which is 1.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.137 C, and the margin is measured, not chosen: 1.137 C of group-conditional forecast error for this hour of day, plus 0.0005 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 0.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.173 C, and the margin is measured, not chosen: 1.173 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.330 C against a 18.0 C limit, so it fails by 0.330 C. It would take a limit 0.330 C higher, or a bound 0.330 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.610 C against a 18.0 C limit, so it fails by 0.610 C. It would take a limit 0.610 C higher, or a bound 0.610 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.050 C against a 18.0 C limit, so it fails by 0.050 C. It would take a limit 0.050 C higher, or a bound 0.050 C tighter, to change this hour.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.501 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.945 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.562 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.116 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.391 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.396 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated 913 -- the agent never calibrates on a day it is scored on
Free cooling delivered 7.00 h/day, i.e. 2,558 h/yr
Chiller-hours avoided +348.0 h/yr against a tuned reactive incumbent
12-hour plan stability 94.3 % of 21,780 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint	+107.2 h/yr
+ switch budget 2, min dwell 3 h	+112.4 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6)	+118.8 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast)	+348.0 h/yr
+ unanchored, 4 measured FG offsets rotated	-87.6 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power	156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity	8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided	54,332 kWh per MW of IT load per year
Value	\$4,738 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

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cd INTAKE-ARBITER/src && python run_all.py
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That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.